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When Left Ventricular Dysfunction Predates Chemotherapy: Severe Cardiomyopathy After Limited Anthracycline Exposure

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Abbreviations

- CMR = cardiac magnetic resonance
- CTRCD = cancer therapy–related cardiac dysfunction
- DCM = dilated cardiomyopathy
- DLBCL = diffuse large B-cell lymphoma
- GDMT = guideline-directed medical therapy
- GLS = global longitudinal strain
- LGE = late gadolinium enhancement
- LVEF = left ventricular ejection fraction
- MUGA = multigated acquisition
- NT-proBNP = N-terminal pro–B-type natriuretic peptide

Abstract

Background: Heart failure after cancer therapy is often attributed to that therapy. Attribution fails when ventricular dysfunction predates treatment.

Case Summary: A 64-year-old woman treated for colorectal diffuse large B-cell lymphoma had a pre-chemotherapy multigated acquisition scan showing left ventricular ejection fraction (LVEF) 43%. Doxorubicin was replaced with etoposide after one cycle. Thirteen months after completing therapy, following bowel surgery, recurrent abscesses, and malnutrition, she presented with LVEF below 10%, mid-wall septal fibrosis on cardiac magnetic resonance, and no coronary disease. Within 3 months of guideline-directed therapy, LVEF recovered to 38% to 40%, within her pre-treatment range.

Discussion: Baseline imaging reframed the case from anthracycline cardiotoxicity to reversible decompensation of pre-existing dysfunction. The post-cycle value near 45% lies within cross-modality measurement variability; recovery to the pre-treatment range argues against fixed end-stage disease.

Take-Home Messages: Reconstruct baseline ventricular function before attributing heart failure to cancer therapy, and compare serial measurements only within the same imaging modality.

History of Presentation

A 64-year-old woman presented with 5 days of progressive exertional dyspnea, orthopnea, and lower-extremity edema. This was 13 months after curative-intent immunochemotherapy for lymphoma and 2 months after a prolonged postoperative course.

She was afebrile and normotensive with persistent sinus tachycardia, elevated jugular venous pressure, bibasilar crackles, and bilateral pitting edema, without hypoperfusion. NT-proBNP was 25,535 pg/mL. High-sensitivity troponin T was flat at 42 to 44 ng/L, lactate was 1.1 mmol/L, and creatinine was 0.56 mg/dL.

Past Medical History

Epstein-Barr virus–positive DLBCL of the rectosigmoid colon had been diagnosed 19 months earlier, with bowel, mesenteric, and splenic involvement complicated by contained perforation and a drained pelvic abscess. She received rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP); doxorubicin was then replaced with etoposide (R-CEOP) because of reduced LVEF. Six cycles were completed. Doxorubicin exposure was reportedly limited to the first cycle (approximately 50 mg/m² at standard dosing); the treating center's administration record is pending.

A chronic enterocutaneous fistula developed after drain removal and required takedown with small-bowel resection. The postoperative course included recurrent intra-abdominal collections, an anterior mediastinal collection requiring drainage, malnutrition, and severe deconditioning.

Other history included Graves disease treated by remote thyroidectomy, on levothyroxine, and premature ventricular complexes noted before chemotherapy.

Differential Diagnosis

Considerations included cancer therapy–related cardiac dysfunction (CTRCD) from limited anthracycline exposure; progression of a pre-existing, possibly genetic, dilated cardiomyopathy (DCM); inflammatory and catabolic injury from prolonged sepsis and malnutrition, including sepsis-induced myocardial dysfunction; tachycardia-mediated cardiomyopathy; thyroid-related cardiomyopathy; ischemic cardiomyopathy; and myocarditis.

Investigations

The decisive investigation preceded chemotherapy. A MUGA scan obtained for cardiac screening before treatment showed LVEF 43% with a mildly enlarged left ventricle and no regional wall-motion abnormality. Oncology documentation subsequently recorded LVEF approximately 45% after the first R-CHOP cycle, which prompted the etoposide substitution. No pre-treatment strain imaging or biomarkers were available, and no ventricular measurement was recorded in the 17 months between then and presentation (Figure 1).

At presentation, echocardiography showed severe left ventricular dilation with LVEF below 10%, GLS −3%, severe functional mitral and tricuspid regurgitation, and an apical echodensity suspicious for thrombus. CMR showed left ventricular end-diastolic volume index 151 mL/m², LVEF 7% to 10%, right ventricular ejection fraction 33%, and mid-wall septal LGE without increased myocardial T2 signal (Figure 2). Prominent trabeculation was interpreted as remodeling in advanced DCM rather than noncompaction. No definite intracardiac thrombus was seen. Coronary computed tomography angiography showed no plaque or stenosis, with a calcium score of zero.

Computed tomography of the chest showed a segmental pulmonary embolism without right-heart strain, pulmonary edema, and bilateral pleural effusions. A catheter-associated right upper-extremity deep venous thrombosis had been identified 9 days earlier during peripherally inserted central catheter placement. Thyrotropin ranged from 1.65 to 5.38 mIU/L with normal free thyroxine (1.33 ng/dL), excluding thyrotoxicosis. Hemoglobin was 9 to 11 g/dL; creatinine was normal.

Management

Treatment comprised intravenous furosemide followed by initiation and titration of GDMT with sacubitril/valsartan, metoprolol succinate, spironolactone, and dapagliflozin. Persistent sinus tachycardia with low-output physiology was treated with digoxin and ivabradine. Apixaban was given for venous thromboembolism and the suspected left ventricular thrombus. A wearable cardioverter-defibrillator was provided pending reassessment of ventricular function, and advanced heart failure follow-up was arranged. Cardiomyopathy genetic testing and right-heart catheterization were recommended at discharge; neither had been done at 8 months.

Outcome and Follow-Up

Congestion improved before discharge. NT-proBNP fell from 25,535 to 1,337 pg/mL, then to 659 pg/mL, and measured 972 pg/mL at 7 weeks (Figure 1). Echocardiography at 3 months showed LVEF 38% by biplane method (40% by visual estimate), GLS −13%, a normal left ventricular cavity size, no mitral regurgitation, and normal right ventricular function; the wearable defibrillator was discontinued. At 5 months she was New York Heart Association class II, euvolemic, with weight recovered from 57 to 73 kg. She had stopped metoprolol, ivabradine, and furosemide for low blood pressure and continued sacubitril/valsartan, spironolactone, dapagliflozin, digoxin, and apixaban. Iron studies at 3 months showed transferrin saturation 12% and ferritin 119 ng/mL; intravenous iron had not been given. Post-treatment positron emission tomography at month 14 showed complete metabolic response (Deauville 1); a surveillance study at month 21 showed hypermetabolic peritoneal nodules (Deauville 5) suspicious for recurrence, unconfirmed in the available record.

Discussion

The pre-treatment LVEF of 43% changes the causal question. Under the Heart Failure Association and International Cardio-Oncology Society baseline proforma, a baseline LVEF below 50% places a patient in the high-risk category before any anthracycline is given (1). For that category, the 2022 European Society of Cardiology (ESC) cardio-oncology guideline recommends cardiology referral before treatment (Class I), consideration of dexrazoxane, liposomal anthracyclines, or renin-angiotensin and beta-blocker therapy (Class IIa), and echocardiographic surveillance during and after therapy — not a 17-month gap (2). Had the baseline result informed those decisions, there may have been an opportunity to initiate heart-failure therapy before oncologic treatment. That gap is more instructive than assigning fault: substituting etoposide preserved curative intent, and R-CEOP retains curative potential when anthracyclines are contraindicated (3).

The apparent decline from 43% to 45% points in the wrong direction. Test–retest variability for two-dimensional echocardiographic LVEF exceeds 10 percentage points, versus about 6 for three-dimensional echocardiography,(4) and MUGA and echocardiographic values are not interchangeable. A 2-point difference across modalities is noise; the likeliest interpretation is recognition of dysfunction already present, not an early treatment effect.

Anthracycline contribution remains possible and unproven. In a large anthracycline-treated cohort, 98% of detected cardiotoxicity occurred within the first year,(5) whereas this patient decompensated 13 months after her final cycle during a period of surgery, recurrent infection, and malnutrition. Sepsis itself depresses the myocardium: profound but reversible ejection-fraction depression, typically resolving within 1 to 2 weeks of recovery, is well described in septic shock and can be superimposed on pre-existing dysfunction (Supplemental Discussion). Imaging establishes a nonischemic phenotype without identifying its cause: LGE is present in only about 10% of patients evaluated for anthracycline or trastuzumab cardiotoxicity and usually has an alternative explanation (6). Absent T2 signal elevation argues against active inflammation at the time of imaging but cannot exclude earlier injury.

The recovery is the most informative finding. Within 3 months, LVEF returned to 38% to 40% with normalization of cavity size and resolution of functional mitral regurgitation, which is within her pre-treatment range, not a new normal. That trajectory indicates a large reversible component on a fixed substrate: sepsis, catabolic stress, tachycardia, and volume overload drove the ventricle from 43% to below 10%, and their reversal, with GDMT, brought it back. Recovery occurred although the beta-blocker and ivabradine were stopped for hypotension, pointing to reversal of precipitants rather than any single drug. Mid-wall LGE marks adverse remodeling and arrhythmic risk in DCM (7); the follow-up echocardiographic findings and the mid-wall LGE on CMR are consistent with a pre-existing cardiomyopathy rather than fully recovered myocardium. She now meets the definition of heart failure with improved ejection fraction; in recovered dilated cardiomyopathy, stopping medical therapy can be associated with decline in systolic function (8), so long-term maximally tolerated GDMT remains recommended, with re-challenge of the discontinued agents as blood pressure allows.

Two considerations follow. First, an unrecognized genetic substrate deserves evaluation: 7.5% of patients with cancer therapy–induced cardiomyopathy were found to have truncating titin variants, versus 1.1% of cancer controls,(9) and genetic testing with family screening is recommended in DCM (7); a mildly dilated, hypokinetic ventricle at 63 before any exposure is how many genetic cardiomyopathies first declare themselves. Second, the prominent trabeculation should not be over-read; hypertrabeculation is now regarded as a phenotypic trait rather than a distinct cardiomyopathy (7). Meanwhile the reversible amplifiers remain actionable: transferrin saturation

of 12% with ferritin 119 ng/mL meets the heart failure definition of iron deficiency, for which intravenous iron carries a Class I recommendation for symptoms and quality of life (10). Ferritin rises with inflammation, so transferrin saturation is the more reliable signal here. Thyroid studies did not support thyrotoxicosis, and a remote history of Graves disease does not establish a thyroid-mediated cardiomyopathy.

Conclusions

Severe decompensation occurred on a pre-existing nonischemic cardiomyopathic substrate after limited anthracycline exposure and major inflammatory–catabolic stress. Serial imaging showed substantial reverse remodeling to the pre-treatment range, and CMR supported secondary hypertrabeculation rather than primary noncompaction. Reconstructing baseline function prevents overattribution to anthracyclines, keeps a possible genetic substrate and reversible precipitants in view, and locates the failure point where it lay here: the baseline result was in the chart, but it never reached the treatment decisions.

Take-Home Messages

Baseline ventricular function must be reconstructed, and serial values compared within a single imaging modality, before heart failure is attributed to cancer therapy. When dysfunction predates treatment, the differential should stay open to a pre-existing cardiomyopathy and to reversible precipitants such as infection, catabolic stress, tachycardia, and iron deficiency, whose reversal can return the ventricle toward its baseline.

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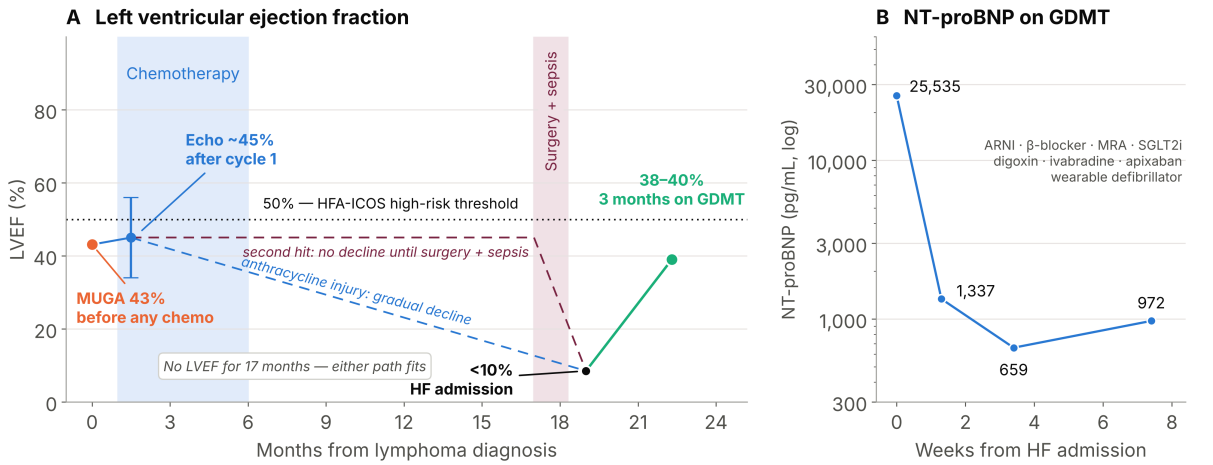
Table 1. Chronology of oncologic, surgical, and cardiac events

Relative timing	Event	Objective data	Cardiac relevance
Month 0	DLBCL diagnosed after weight loss, abdominal pain, anemia	EBV-positive DLBCL of rectosigmoid colon; staging PET with bowel, mesenteric, and splenic disease; contained perforation with pelvic abscess drained	Systemic inflammatory burden concurrent with planned therapy
Month 0 (before cycle 1)	Pre-treatment cardiac screening	MUGA LVEF 43%; mildly enlarged LV; no regional wall-motion abnormality	High-risk category before any anthracycline; the pivotal finding

Relative timing	Event	Objective data	Cardiac relevance
Months 1–6	Immunochemotherapy	R-CHOP started; LVEF approximately 45% recorded after cycle 1; etoposide substituted for doxorubicin (R-CEOP); 6 cycles completed. Doxorubicin limited to cycle 1 (~50 mg/m²)	Change in regimen limited exposure; 2-point difference is within cross-modality variability
Months 7–17	Chronic enterocutaneous fistula after drain removal	Recurrent presentations; complete metabolic response on post-therapy PET (month 14)	Chronic inflammation and nutritional depletion
Month 17	Fistula takedown with small-bowel resection	Preoperative exercise tolerance reported as good	Major physiologic stressor
Months 17–18	Prolonged postoperative course	Recurrent intra-abdominal collections; anterior mediastinal collection drained; malnutrition; deconditioning	Sustained inflammatory and catabolic state
Month 19	Decompensated heart failure	NT-proBNP 25,535 pg/mL; segmental PE (catheter-associated upper-extremity DVT 9 days earlier); echocardiographic LVEF <10% with GLS −3%; 4+ MR, 3+ TR; apical echodensity	End-stage phenotype with thromboembolism
Month 19	Advanced imaging	CMR LVEDVI 151 mL/m², LVEF 7%–10%, RVEF 33%, mid-wall septal LGE, no T2 elevation; coronary CTA without plaque, calcium score 0	Nonischemic fibrotic substrate, not modality-specific for anthracycline injury
Months 19–21	Early follow-up	NT-proBNP 1,337 → 659 → 972 pg/mL; creatinine 0.80 mg/dL	Biomarker response on GDMT
Month 21	Surveillance PET	Hypermetabolic peritoneal nodules, Deauville 5, suspicious for recurrence; unconfirmed	Any salvage therapy would now be planned around her cardiac status
Month 22	3-month echocardiogram	LVEF 38% (biplane), GLS −13%; normal LV cavity size; no MR; normal RV; wearable defibrillator discontinued	Return to the pre-treatment range: heart failure with improved EF
Month 24	Clinic follow-up	NYHA II; weight 57 → 73 kg, euvolemic; metoprolol, ivabradine, furosemide self-discontinued for hypotension; iron deficiency (TSAT 12%, ferritin 119) untreated; genetic testing not yet done	Persistent substrate; GDMT to continue indefinitely

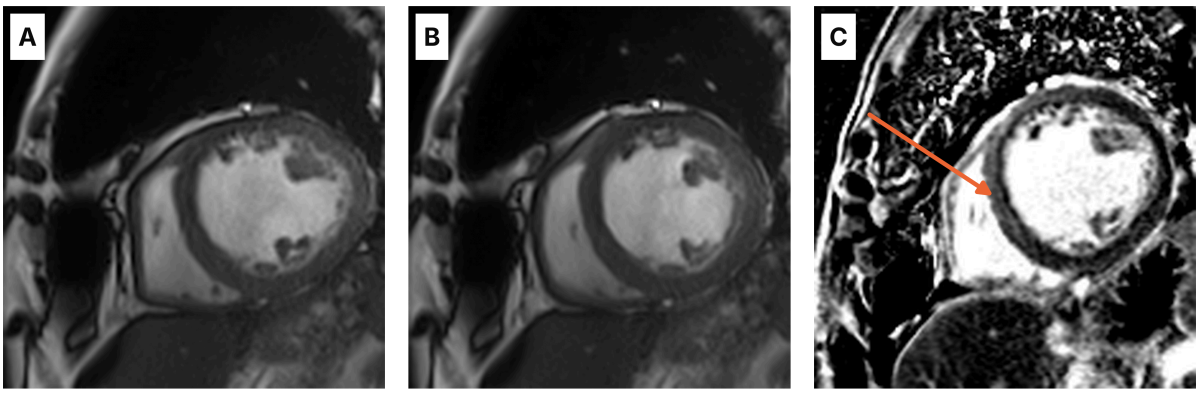
CMR = cardiac magnetic resonance; CTA = computed tomography angiography; DLBCL = diffuse large B-cell lymphoma; DVT = deep venous thrombosis; EBV = Epstein-Barr virus; GLS = global longitudinal strain; LGE = late gadolinium enhancement; LV = left ventricular; LVEDVI = LV end-diastolic volume index; LVEF = LV ejection fraction; MR = mitral regurgitation; MUGA = multigated acquisition; NT-proBNP = N-terminal pro–B-type natriuretic peptide; NYHA = New York Heart Association; PE = pulmonary embolism; PET = positron emission tomography; RV = right ventricle; TR = tricuspid regurgitation; TSAT = transferrin saturation.

Figure 1. Ventricular function and biomarker course (Visual Summary)



(A) Left ventricular ejection fraction across oncologic and surgical events: the pre-treatment multigated acquisition value of 43% (orange); the echocardiographic value near 45% after cycle 1, with an error bar spanning the ±11–percentage point test–retest variability of two-dimensional echocardiography (blue); the nadir below 10% at the heart-failure admission (black); and recovery to 38% to 40% at 3 months of guideline-directed therapy (green), within the pre-treatment range. Dashed paths span the 17-month interval in which no ventricular measurement was made and depict the two competing explanations: gradual anthracycline injury (blue) versus no decline until the surgical–septic second hit (maroon). Shaded bands mark chemotherapy (blue) and the perioperative–septic period (maroon); the dotted black line marks the 50% threshold below which baseline risk is classified high. **(B)** NT-proBNP fell more than 95% on guideline-directed therapy and plateaued near 1,000 pg/mL.

Figure 2. Cardiac magnetic resonance at the heart-failure admission (month 19)



One mid-ventricular short-axis slice, 13 months after the final chemotherapy cycle. **(A, B)** Balanced steady-state free-precession cine at end-diastole and end-systole: a dilated, thin-walled left ventricle with minimal systolic excursion (LVEF 7% to 10%; left ventricular end-diastolic volume index 151 mL/m²). **(C)** The same slice on phase-sensitive inversion-recovery late gadolinium enhancement: linear mid-wall septal enhancement (arrow), without subendocardial coronary-territory scar, without increased T2 signal, and without definite thrombus — a nonischemic fibrotic substrate. Trabeculation was read as secondary to dilation rather than noncompaction.

Patient Consent

Written informed consent for publication was obtained from the patient.

Supplemental Discussion (online supplement)

Classification of cancer therapy–related cardiac dysfunction. CTRCD has classically been divided into two types (S1). Type I, typified by anthracyclines, is dose-dependent, produces ultrastructural myocyte injury, and is frequently irreversible. Type II, typified by trastuzumab and other HER2- or VEGF-directed agents, is not dose-related, lacks ultrastructural change, and is largely reversible on withdrawal (S1, S2). This patient's course fits neither category cleanly: dysfunction preceded any exposure, cumulative doxorubicin was reportedly about 50 mg/m², and the collapse reversed to her pre-treatment range within 3 months — a trajectory atypical for Type I injury at that dose.

Sepsis-induced myocardial dysfunction. In the original radionuclide description of septic shock, half of patients had an ejection fraction below 40% in the acute phase; survivors had a mean initial ejection fraction of 32% with ventricular dilation, and ejection fraction and volumes normalized by

about 10 days after shock onset (S3). Sepsis-induced myocardial dysfunction is global rather than regional, is distinct from stress (takotsubo) cardiomyopathy, can be superimposed on pre-existing ventricular dysfunction, and is highly reversible with resolution of sepsis. In this case, fistula takedown, recurrent intra-abdominal and mediastinal collections, and malnutrition immediately preceded the decompensation — making sepsis-related myocardial depression, on top of the pre-existing substrate, the most economical explanation for the depth of the nadir and the speed of recovery.

Supplemental references

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